Example 7.25

Evaluate $\int_0^1 \int_0^1 16abxydx\,dy$

Solution:

$$\int_0^1 \int_0^1 16abxydx\,dy$$

$$= 16ab \int_0^1 \left[\int_0^1 xydx \right] dy$$

$$= 16ab \int_0^1 \left[\frac{x^2}{2} y \right]_0^1 dy$$

$$= 16ab \int_0^1 \left[\frac{y}{2} \right] dy$$

$$= 8ab \left[\frac{y^2}{2} \right]_0^1 = 4ab[1-0]$$

$$= 4ab$$

Points to Remember

- Fundamental Principle of multiplication on counting: If an operation can be performed in m ways and if another operation in n ways independent of the first, then the number of ways of performing both the operations simultaneously in $m \times n$ ways.
- Factorial: The consecutive product of first n natural numbers is known as factorial n and is denoted as n ! or $\underline{|n}$
- Permutations : Thus the number of arrangements that can be formed out of n things taken r at a time is known as the number of permutation of n things taken r at a time and is denoted as nP_r; $nP_r = n(n-1)(n-2) - - - - [n-(r-1)] = \frac{n!}{(n-r)!}$
- Combinations : The number of combination of n different things, taken r at a time is given by $nC_r = \frac{np_r}{r!} = \frac{n!}{(n-r)!r!}$, $nC_r = nC_{n-r}$
- Binomial series : For any natural number n

  $(x+a)^n = nc_0 x^n a^0 + nc_1 x^n - 1_a + nc_2 x^{n-2} a^2 + \ldots + nc_r x^{n-r} a^r + \ldots + nc_n a^n$

  For any rational number other than positive integer

  $(1+x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$, provided $|x| < 1$.
- Exponential series : For all real values of x, $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$ is known as exponential series, Here, e= 2.718282